Updated requirements SWE NUT

The costumer requires an nutrition app in order to provide excessive assistance to clients with creating healthy meal plans and support to create a healthy habits.

The customer would like to see these main features:

- Reminders for food consumption
- Chats with nutritionists
- Meal search
- Meal plan generator based on data
- Personal accounts with user's data
- Shopping list generator
- Meal generator based on food stocks

1. Reminders – Konstantin

Problem:

Users sometimes forget to take their meals or drink water, which makes it hard to follow their meal plan and reach their health goals.

Future:

1. Upon opening the application, the user will see a display showing only the name and the date/time of the next closest scheduled consumption.
2. The user can click on this specific display to see the full details of that reminder.
3. When a reminder appears, the user can choose to either "confirm" or "decline" the consumption. If the user selects "confirm", the nutritional data is automatically added to the daily progress log and the ingredients are removed from the pantry. If the user selects "decline", no data is changed, and the reminder disappears.

4. Whenever a new daily meal plan is generated, the system must automatically create reminders for breakfast, lunch, and dinner based on the user's chosen meal times.
5. There is a "Reminders" button that, when clicked, shows a list containing the name, time, and date of every scheduled reminder.
6. The user can create a new reminder or edit/delete an existing one. To save a reminder, the following rules apply:

4.1. The Name field can only accept 50 alphanumeric characters.

4. 3. The user must pick a Time and a Date for the reminder appear.

4.4. The user must select a Recurrence from a list of "daily", "weekly", OR "monthly".

4.8. The "Submit" button remains disabled until all fields are filled with valid data.

4.7. Fallbacks:

If the user clicks "Delete," a confirmation window will appear; the reminder is only deleted if the user clicks "Yes", or the user can cancel this process.

5. If no reminders exist, the text "No reminders scheduled" will be displayed

2. Logging and daily/weekly reports – Claudiu (cross team, VibeCoders)

Problem: Users cannot easily track their nutritional intake or physical activity over time, making it difficult to visualize if they are meeting the targets set by their generated meal plans.

Future:

1. There is a button in the header labeled "Progress".
2. This section displays the following five values:
    a. Total calories consumed and total calories burnt.
    b. Total grams of protein, carbohydrates, and fats consumed.
3. The information is divided into two areas:

    a. The left area displays only the data for the current day.
    b. The right area displays the combined data for the current week up to the present moment.
4. Conditions:
    a. These values are ONLY displayed IF 24 hours have passed since the user's first meal plan was created.
    b. All weekly progress data MUST automatically reset to zero at the start of every new week.
5. Under this info the user will be presented with the option to log a meal, an user will be able to search for a meal and then submit it as consumed. After this the meal plan will be regenerated to fit their goal again.

6. Fallbacks:
    a. If 24 hours have not yet passed since the first meal plan was generated, the left area will display only the text: "Not enough information to display".

3. Pantry and Food Stock Management

Problem

Users lose track of the ingredients they already own, which leads to either purchasing unnecessary duplicates or being unable to verify if they can cook a planned meal .

Future

1. There is a symbol labeled "pantry" that displays a list of all food items currently in the user's possession .
2. For each item, the user can see the name and the current weight in grams.
3. A user can remove any item from this list by selecting an "x" symbol next to the item name.
4. The user can manually add ingredients by selecting a cross symbol. to add an item, the user must provide a name and a weight .
5. The weight provided must be only a positive whole number.
6. The pantry list updates automatically:

- ingredients are added only when a user marks an item as purchased on their shopping list.
- ingredients are removed only when a user logs a specific meal as consumed.

7. **fallbacks:**
   - if the pantry is empty, the screen displays only the text: "your pantry is empty; add ingredients to start".
   - if a user enters a weight of zero or less, the text "please enter a quantity greater than zero" will appear.

4. Smart Shopping List Generation

Problem

Users are not provided with a specific list of missing items needed for their meal plan, which causes them to lack the necessary ingredients for healthy cooking.

Future

1. There is a shopping cart symbol that displays a list of ingredients required for the user's current meal plan only if those items are not already present in the pantry .
2. For every item, the user has two choices:
   - selecting a checkmark removes the item from the list and adds that quantity to the pantry.
   - selecting an "x" mark removes the item from the list only.
3. Any removal of an item from the shopping list requires a final confirmation from the user.
4. The user can manually add items by providing a name and selecting only one ingredient from a provided list of suggestions .
5. Fallbacks:
   - if all ingredients for a plan are already owned, the list displays only the text: "you already have everything you need".
   - if a manual search for an item yields zero results, the text "no matching ingredients found" will be shown.

5. Instant Meal Substitution (Pantry-First)

Problem

Users occasionally want to cook a meal using only what they currently have available without following the pre-generated plan or going to the store .

Future

1. Within the meal plan area, there is an option to "substitute a meal using pantry items".
2. The user must choose which specific meal they wish to replace: "breakfast", "lunch", or "dinner".
3. Once a meal type is chosen, the system displays only those recipes where 100% of the required ingredients are currently in the user's pantry.
4. After the user selects and confirms a substitute meal, the daily meal plan is updated to show the new choice .
5. The ingredients used in the substituted meal are automatically removed from the pantry when the user confirmed consumption.
6. **fallbacks:**
    a. if no meals can be created using the current pantry stock for the chosen meal type, the text "no meals match your current stock for this category" will be shown.
    b. the user can cancel the substitution process at any time, which keeps the original meal plan unchanged.

6. Favorite meals - Razvan

Problem

The application currently generates meal plans based only on physical data such as allergies and health goals . there is no method for a user to prioritize meals they personally enjoy, which can lead to meal plans that do not suit the user's tastes.

Future

1. Within the meal search area, every meal displays a grey star symbol to the right of the name.
2. A user can select the grey star to mark a meal as a favorite.
3. The star symbol changes from grey to yellow only after the user selects it.
4. A user can select the yellow star to remove a meal from their favorites, which changes the symbol back to grey.
5. If a user has at least one favorite meal, the system must prioritize including these meals in the next daily meal plan generation.
6. A favorite meal can only appear in a generated plan if it has not been included in a plan during the previous three days
7. Fallbacks**:**

- if no meals have been marked as favorites, the system will generate plans using only the user's health data and goals.
- if all favorite meals have appeared in a plan within the last three days, the system will generate a plan using only standard health criteria until the three-day limit has passed.
- if a meal is no longer available in the system, it is automatically removed from the user's favorites list.

7 Automatically generate meal plans based on user given data – Mirko

Problem

Currently, users are not provided with meal plans that suit their personal physical needs and health goals, making it difficult to maintain a healthy diet.

Future

1. During the sign-up process, a user with role “user” (not “nutritionist”) must provide personal data before their profile is finalized.
2. The user must provide the following information:
   - weight in kilograms, using only positive whole numbers.
   - height in centimeters, using only positive whole numbers.
   - birthdate a valid date,
   - gender, by selecting either "male" or "female".
   - a health goal, by selecting from "bulk", "cut", "maintenance", or "well-being".

3. Based on the height and weight provided, the system displays a recommended goal for the user.
4. The recommended goal can be acquired from the formula:
   - BMI= weight/(height/100)^2.
   - **< 18.5 → recommend "bulk"**
   - **18.5 – 24.9 → "maintenance"**
   - **25+ → "cut"**

5. The "done" option is available only after all personal information has been provided correctly.
6. Upon completion, the system automatically calculates the user's daily calorie and nutrient needs.
7. Nutrient needs will be obtained from the formula:
   - BMR=10·weight+6.25·height−5·age+5 (for men)
   - BMR=10·weight+6.25·height−5·age−161 (for women)

   And the meal generation will take these into consideration when generating meal plans:

   - **Cut (fat loss):**
     → BMR− 300 to 500 kcal
   - **Bulk (muscle gain):**
     → BMR + 300 to 500 kcal
   - **Maintenance:**
     → BMR
   - **Well-being:**
   - → BMR +/- 100 calories
8. A new meal plan that fits these needs is created every day and is shown in the meal plan area.
9. A user can change their personal data or health goal at any time by selecting the "settings" symbol in the header.
10. Fallbacks
    - if a user enters a value that is not a positive whole number, the text "please enter a valid number" will be shown.
    - if no meal plan can be generated for the specific needs, the text "no meals found matching your current goals" will be displayed.

8. Nutritionist chats –Anisia

Problem

Currently, users do not have the ability to receive professional advice tailored to their specific health needs or their generated meal plans.

Future:

1. There is a chat symbol that, when selected, displays a conversation area for professional advice.
2. The conversation area is organized so that messages from the nutritionist appear only on the left, and messages from the user appear only on the right.
3. The conversation area must automatically display all previous messages exchanged between the user and nutritionists in chronological order.
4. A message will contain the date it was sent at and the username of the person who sent it.
5. To send a message, the user or nutritionist must provide text and select the send option.
6. A message can consist only of alphanumeric characters and must be at most 1000 characters long.
7. Permissions:
   - a standard user can only read or write messages within their own personal conversation area.
   - a nutritionist can read or write messages in any active user conversation.
8. A nutritionist will have access to 2 tabs, “Show all chats” and “Show my chats”.
9. “Show all chats” will show the chats from all the users who have sent at least 1 message in their chat to any nutritionists and “show my chats” will display the chats in which the nutritionist has responded atleast once.
10. If there are any unanswered messages, “nutritionist” should see them first and highlighted.
11. Fallbacks:
    - if a user attempts to send a message longer than 300 characters, the send option will remain disabled and the text "message too long" will be displayed.
    - if no messages have been exchanged yet, the conversation area will display only the text: "start a conversation with our professional nutritionist".

- if the nutritionist list is empty, the screen will display "no active user inquiries at this time".

9. Meal Search with Filtering – Baldean

Problem

Users cannot easily find meals that align with their specific dietary restrictions or allergies, which makes it difficult to follow their personalized meal plans.

Future

1. In the meal search area, there is a symbol labeled "filter".
2. When this symbol is selected, a list of dietary requirements is shown, including "keto", "vegan", "gluten-free", "no nuts,", "lactose-free", “community recipe”.
3. The user can select one or more of these requirements at the same time.
4. After selecting the "apply" option, the area will display only the meals that satisfy every chosen requirement.
5. The user can remove all selected requirements by choosing a "reset" option.
6. A user can select any specific meal from the results to see a detailed display.
7. This detailed display must show the meal name, the image, the total calories, the full list of ingredients.
8. For every ingredient in the list, the user can see both the name and the exact quantity required for the meal.
9. The user can return to the search results only by selecting the "back" symbol.
10. Fallbacks:
    a. if no meals match the chosen requirements, the text "no meals fit this search criteria" will be shown.
    b. if the user selects "apply" without choosing any specific requirements, the area will display all available meals.
    c. if the meal search results cannot be reached, the text "unable to load meals at this time" will appear.

10. Meal plan page

Problem:

Users need to see a meal plan tailored to their specific needs, so that they can achieve their goals

Future:

1. There will be a page called “Meal plan” which users can access
2. On this page will appear the meal plan for the day.
3. A meal plan is a list of 3 meals, Breakfast, Lunch and Dinner
4. An user can mark the meals one by one to signal that they have been eaten for that day.
5. Once this happens the meal’s calories will be added to the daily/weekly logs, any pantry items that were needed for the meal and were present in the user’s pantry will also be removed from the user’s pantry and the marked meal will visibly show it was consumed.

11. User authentication and access

Problem

Users cannot securely save their nutritional progress, physical data, or personalized preferences without a unique personal account, which prevents the application from providing accurate long-term health assistance.

Future

1. There is a symbol labelled as personal account which can provide access to personal data.
2. If a user is not already identified by the system, they can choose to either "log in" or "create an account".
    - to log in, a user must provide their username and their password.
    - to create a new account:
        i. A Username of 30 alphanumeric characters
        ii. a password of maximum 30 special and alphanumeric character and of length of 8 minimum
        iii. By default, role is “user”, but if user wants to register as a nutritionist, he must change role to "nutritionist”.
3. Access to the application features is granted only after the user has successfully finished the login or registration process with valid data.

4. User stays logged in until they choose the "sign out" option in the settings area. On sign out, they will be presented again with the options to either “log in” or “create an account”

12. Adaptive Meal Plan Optimization – (Behavior-Based Adjustment)

Problem

Users do not always follow their generated meal plans exactly. They may skip meals, substitute them, or log different foods, which makes the generated plans less effective over time since they are based only on static data (such as BMR and goals).

Future

The system automatically adjusts future meal plans based on the user’s behavior over time.

Each time a user interacts with meals, the system records:

- meal type (breakfast, lunch, dinner)
- action type (consumed, skipped, substituted, manually_logged)
- meal ID
- date and time

The system analyzes this stored activity data to modify future generated meal plans.

Behavior Rules:

- If a user skips the same meal type at least 2 times within 3 days, the system will reduce the calorie allocation for that meal or replace it with lighter meals.

- If a user frequently substitutes meals using pantry items, the system will prioritize simpler meals with fewer ingredients.
- If a user consistently logs meals different from the generated plan, the system will detect patterns and include similar meals in future plans, as long as they match the user's nutritional goals.
- If a favorite meal is repeatedly consumed, its priority increases in future plans. If it is ignored multiple times, its priority decreases.
- If the user's daily calorie intake is consistently below or above the target for at least 3 days:
    - below target → increase daily calories by up to 100 kcal
    - above target → decrease daily calories by up to 100 kcal